

Internship Progress Report

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Internship Topic: Design and Development of Just-in-Time Immersive Video Coaching to Support CPAP Adherence

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Role: PhD Researcher / Project Supervisor

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1. Project Context

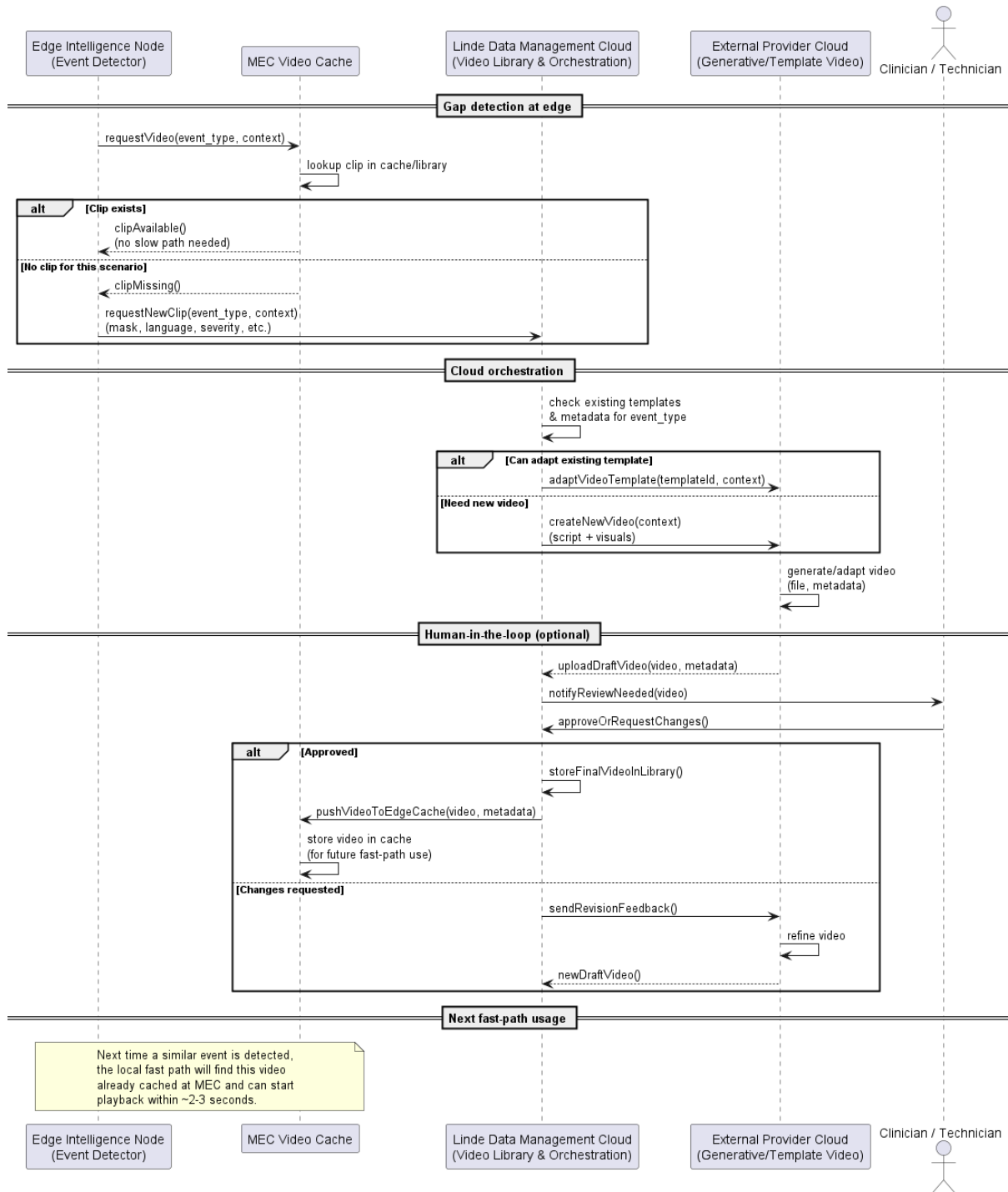
Design and implementation of a 5G Quality on Demand (QoD) enabled dynamic video generation pipeline. The project utilizes Edge Computing (MEC) via Raspberry Pi to detect real-time CPAP telemetry anomalies. Upon detection, it automatically orchestrates the generation and delivery of personalized, medically accurate coaching videos to patients under strict 5G/6G network delivery constraints (<60 seconds).

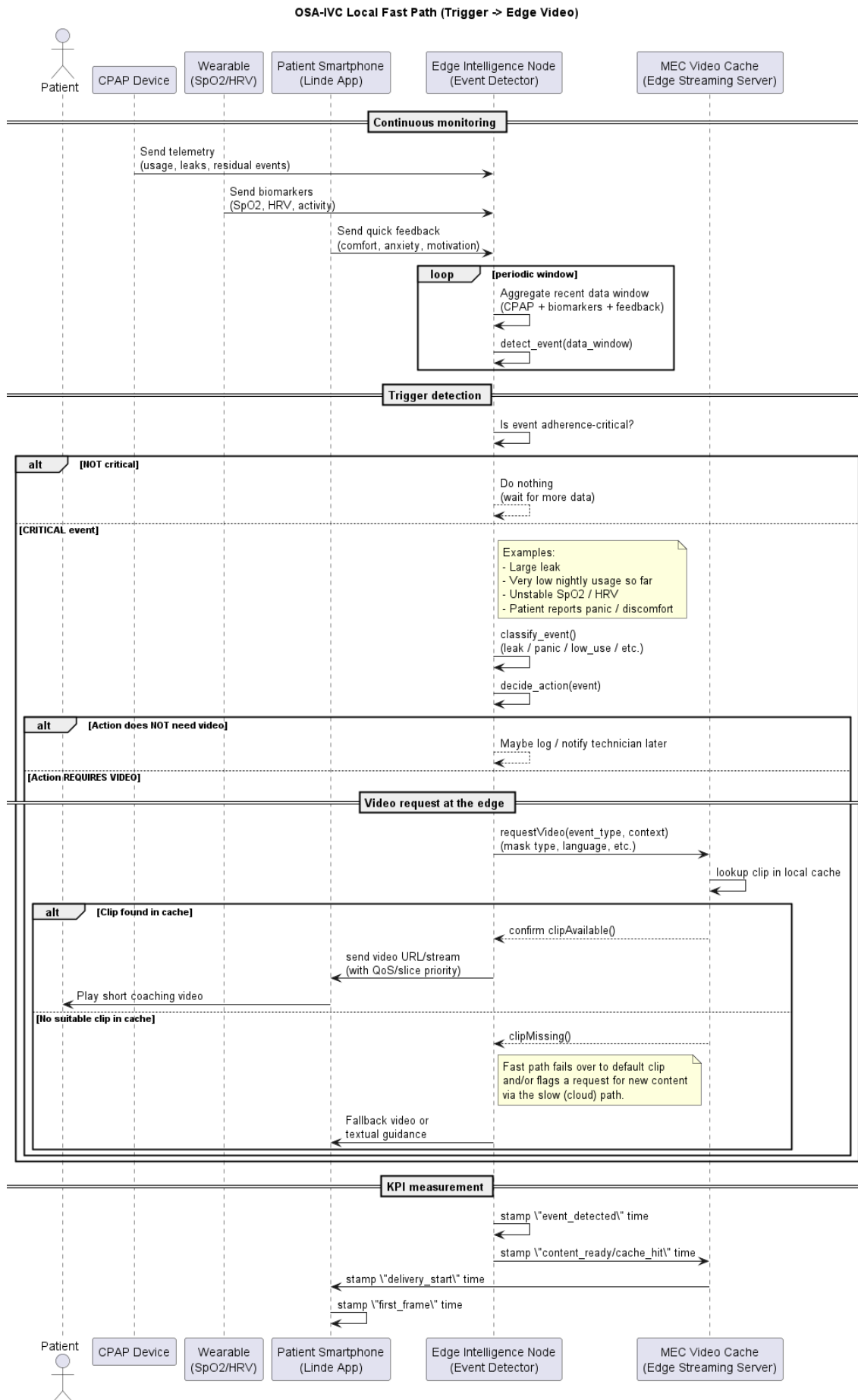
2. Summary of Achievements to Date

During this first month, I laid the foundation for the data ingestion workflow, defined the edge orchestration architecture, and finalized the video generation pipeline.

- **Architecture & Sequence Mapping (Week 1):**
 - Analyzed the 6G Path sequence diagrams to fully map the event-detection logic across the **fast path** (Smartphone → Edge) and **slow path** (Cloud → Edge).
 - Began designing the edge orchestration microservice that will map specific telemetry events to corresponding video coaching actions.

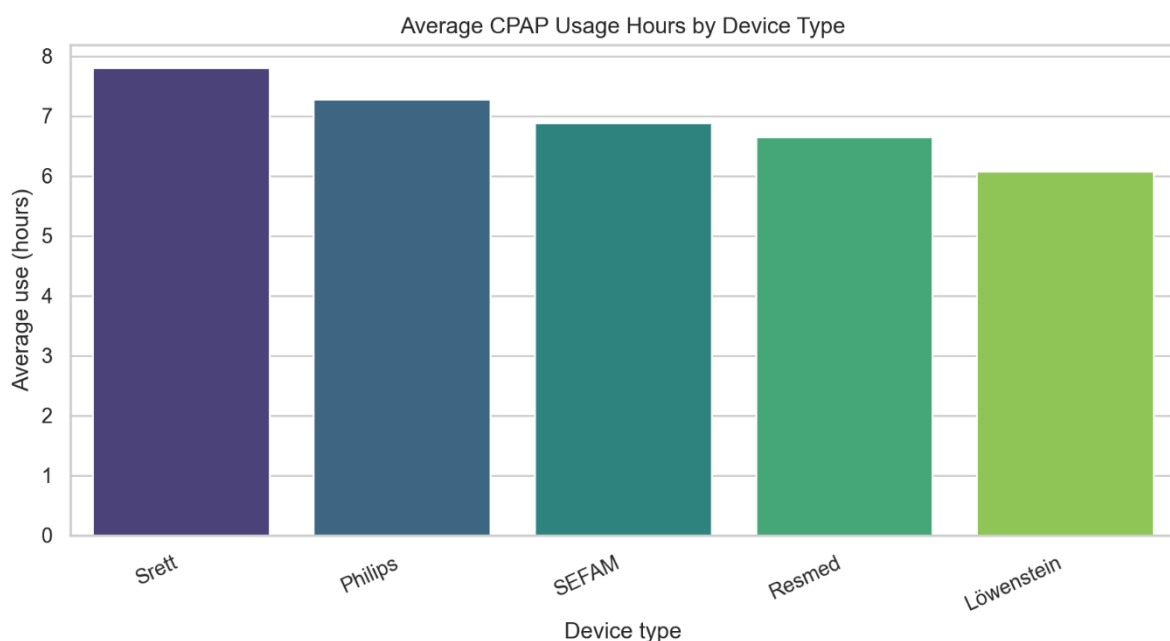
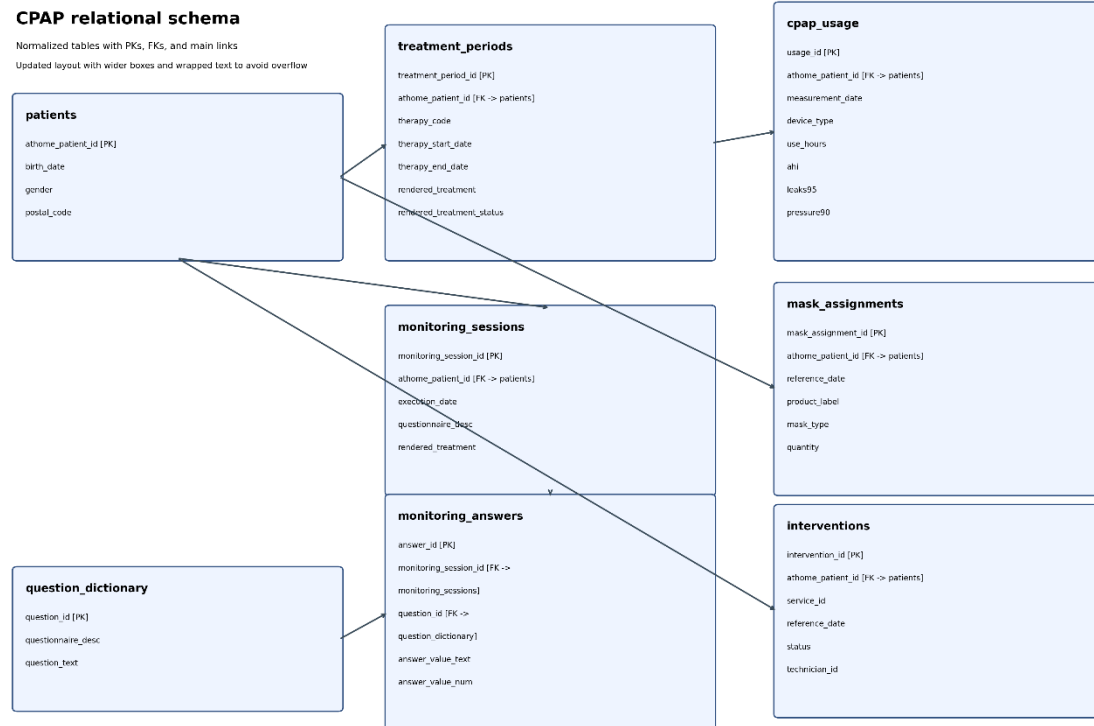
OSA-IVC Global Slow Path (New/Updated Coaching Videos)





- **CPAP Data Pipeline & ETL Development (Week 2):**

- Reviewed raw CPAP CSV files and identified six core data groups: usage, monitoring, masks, patients, treatment periods, and interventions.
- Designed a normalized relational schema with primary/foreign keys and built a complete ETL Jupyter notebook to clean, aggregate, and store this data in a local SQLite database.
- Generated initial data visualizations, prioritizing key metrics such as AHI (Apnea-Hypopnea Index), usage hours, and mask leaks.

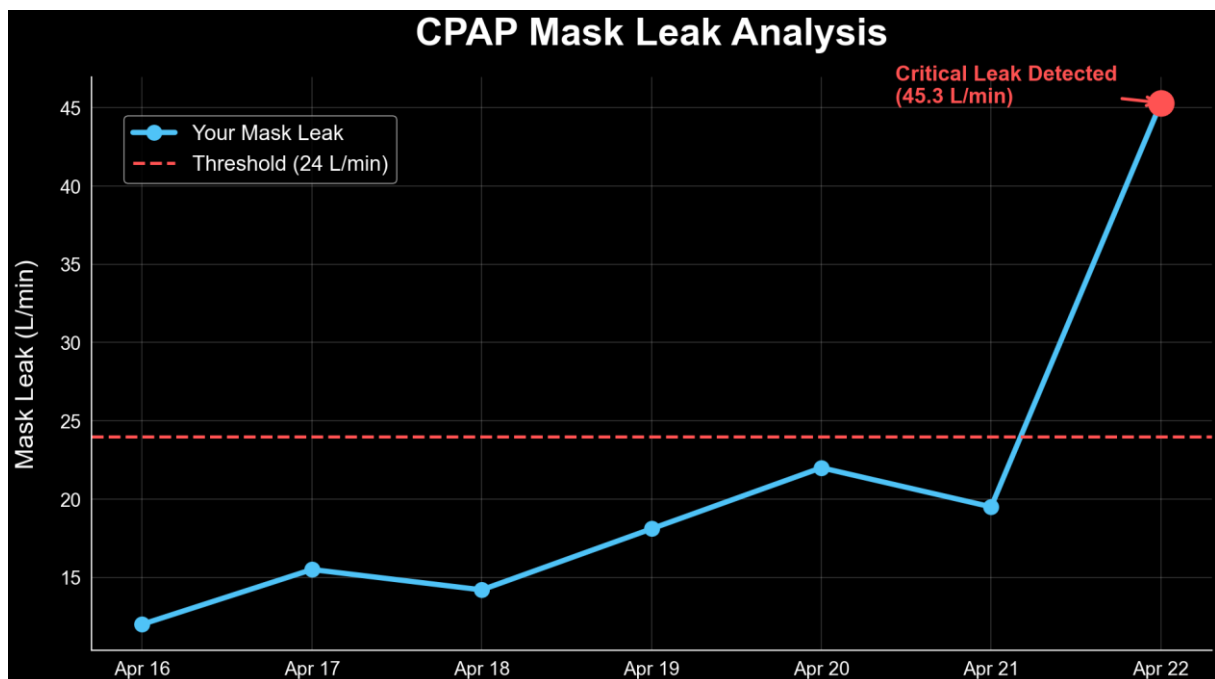


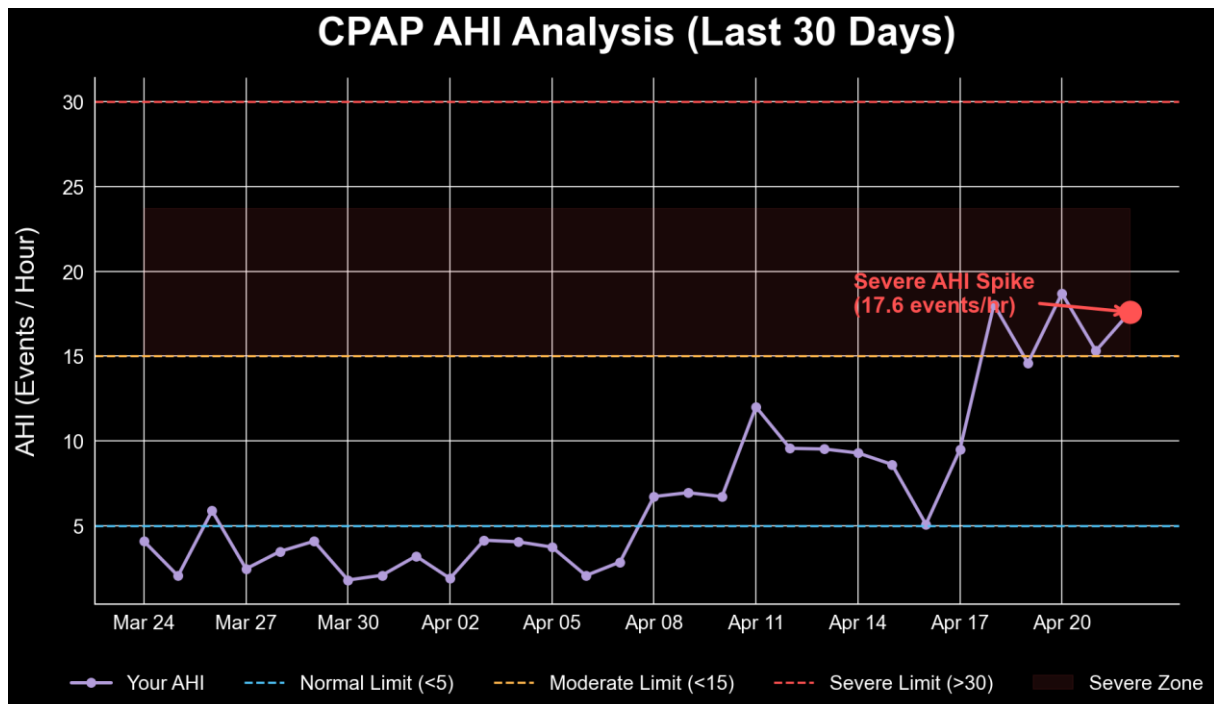
- **Edge Ingestion via Raspberry Pi (Week 3):**

- Set up a secure edge receiver on a Raspberry Pi virtual machine.
- Enabled the SSH server, created the /data/incoming directory, and configured SCP (Secure Copy Protocol).
- Successfully validated the end-to-end file transfer workflow by pushing test CSV telemetry files from a simulated smartphone laptop to the Raspberry Pi over the local network.

- **Video Generation Architecture (Week 4 & 5):**

- Evaluated generative AI APIs (Luma AI vs. Google Veo) by generating initial test clips. Decided on a hybrid architecture to prevent AI from hallucinating medical data.
- Wrote Python (Matplotlib) scripts to generate 100% mathematically accurate, cinematic static charts for priority medical triggers (e.g., AHI spikes, critical mask leaks).
- Built a Python module to programmatically generate HTML5 WebVTT (.vtt) subtitle files in English and French to decouple text generation from the AI video models.





3. Upcoming Schedule (May 2026)

Next month, the focus shifts to deploying the machine learning models to the physical edge hardware and initializing the cloud API access.

- **Week 6:** Convert the existing Keras ML model to TensorFlow Lite (TFLite) to minimize memory usage, then deploy and test inference on the *physical* Raspberry Pi.
- **Week 7:** Integrate the received CSV workflow on the Raspberry Pi: connect the SCP ingestion folder to the TFLite model to trigger immediate local anomaly detection.
- **Week 8:** Set up the cloud video server mapping table and implement a *mock* CAMARA QoD API endpoint to test our network bandwidth reservation logic in software.
- **Week 9:** Use Google Flow and the Vertex AI API to generate the first baseline set of animated videos (Image-to-Video) and sync them with the auto-generated .vtt subtitle tracks.

4. Difficulties Encountered & Resolutions

- **Hardware Availability Delay:** The end-to-end testing of the SCP ingestion workflow was temporarily blocked by the lack of physical edge hardware.

Resolution: I successfully deployed a Raspberry Pi VM on my local machine to validate the SSH and file transfer protocols, ensuring the code is ready to migrate instantly once the physical Pi was provided.

- **Medical Data Hallucination:** Generative AI video models frequently misspell text overlays and warp exact medical numbers.

Resolution: I redesigned the pipeline to decouple data from the AI. Python generates a static chart first, and Vertex AI is only used to animate it and provide native audio. Text is handled entirely via secure .vtt subtitle files.

- **Cloud API Access Requirements:** We need programmatic API access to trigger videos automatically from the Raspberry Pi in under 5 minutes. Consumer Gmail accounts cannot access the Vertex AI developer endpoints.

Resolution: I have submitted a request outlining the exact steps and budget) required to establish a Google Workspace Business Standard account for the lab.